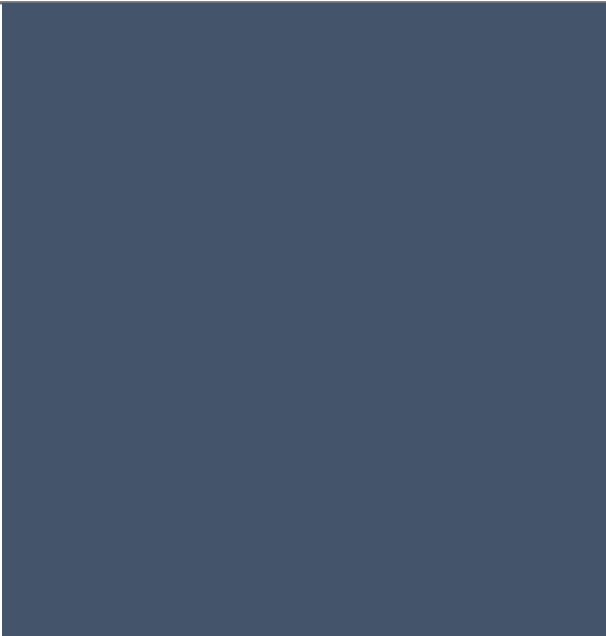




Walamoo Tax System Requirements

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Walamoo Tax Collection and Management System

Requirements

1. Introduction

The Walamoo Governing Council wishes to upgrade its tax information system or TIFS. They have hired your company to develop the user-interface. The Professor has created this document to outline the basic requirements for the system.

Before continuing, read the other related Walamoo documentation.

2. Requirements

Below are the requirements.

1. When the system is first started it must list the total number of records stored in the database, the total amount of tax paid for the year, the total number of countries taxpayers come from, and the total number of cities taxpayers originated from. The user can select an option which opens a window that lists cities per country. In other words, under each country the total number of cities is listed.
2. Data for the system will be provided by an XML file whose format is described later in this document.
3. The user should be able to enter queries as described later in this document.
4. The user should be able to enter threshold and alert values as described later in this document.
5. The user should be able to select the XML file with the data and load it into the system. The system should alert the user whether or not the file was successfully loaded. If the file was not successfully loaded the user should be told why and how to mitigate the issue.
6. The user should be able to reload a new XML file into the system. The reloading operation should remove all previous XML data from the system and replace it with the new XML data.
7. The system should allow the results of all queries to be stored to a text file chosen by the employee. If the file already exists, the user should be asked whether or not to overwrite it. At the top of the text file, the date and time that the file was created should be included. The second line should include the total number of records stored in the text file where a record is considered to be an item that matched the query.
8. The user is required to log in to the system. If the user is unsuccessful after three attempts the system should notify the user that they should speak to the government information systems unit.

9. The system keeps track of when a user logs in and out of the system. Only the manager can view this information.
10. The system should provide the option to translate the generated report to DFL (Digital Frequency Language) and displayed for review. It can be saved to a file if required. See the *Walamoo Digital Frequency Language* document for further details.
11. The Walamoo Disabilities Protection Act requires the software to meet specific digital accessibility requirements. For this software, the user can set the colour of the text to black, blue or green. When the user logs in, the system asks the user to select one of the three colours. At this point, the user can choose to make the colour selection permanent for that user and hence remove the need for the colour choice to be made when logging in.
12. Queries come in two forms.
 - **Guided:** Commonly made queries are done via the completion of fields and the selection of specific attributes via a query builder window.
 - **Ad hoc:** Queries are created via a command language. The query is checked to ensure it is syntactically correct before its execution. If it is incorrect, the user is informed. See below for further details.

3. XML Data

The data for TIFS will be stored as an XML file with the following structure.

```
<tifs>
  <taxpayer id="2452">
    <company>Food Fashion Inc</company>
    <street>Berley Street</street>
    <city>Bridgetown</city>
    <country>Barbados</country>
    <tax>201220</tax>
  </taxpayer>

  <taxpayer id="4251">
    <company>Who Wants Dairy Inc</company>
    <street>Tank Street</street>
    <city>Oistins</city>
    <country>Barbados</country>
    <tax>451909</tax>
  </taxpayer>
</tifs>
```

The meaning of each tag/attribute is given in the table below.

Tag/Attribute	Description
taxpayer	The start of an XML record for the taxpayer.
company	The name of the company.

id	The company tax id.
street	The name of the street that the taxpayer lives on.
city	The name of the city where the street is located.
country	The name of the country where the city is located.
tax	Total taxes paid by the company. A company can pay in installments and so there can be more than one record per company.

Table: Description of Each XML Tag and Attribute

One of these records is created each time a company makes a payment.

4. Monthly Scenarios

The system allows the user to run a monthly scenario. When this option is selected, the user is prompted to enter a month. Once entered, the system checks the set thresholds and alerts against the set month and generates the relevant information. See below for further details.

5. Thresholds

Thresholds are values that define specific data points that must be adhered to. The values for the required thresholds must be entered in order for the system to operate. If any of those values has not been entered, the system must prompt the user to enter them.

Thresholds are checked when a scenario is run.

Threshold	Description	Required	Action
minimum_tax	The minimum tax each company must pay. A company may have to pay more taxes than the minimum based on their total annual earnings.	Yes	Lists the name of each company whose total combined taxes falls under this value.
deadline_month	The month by which all taxes must be paid. The taxes must be paid by the first of that month.	Yes	Checks this month with that on the computer. If the month has passed, run the minimum_tax threshold check.
minimum_revenue	The minimum total taxes that must be collected.	Yes	States whether the total taxes paid has exceeded this value.

minimum_companies	The minimum number of companies that must pay taxes at or above the minimum tax value.	No	States whether or not the total number of companies that have paid at or above the minimum tax threshold, equals or has exceeded this value.
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Table 1: Threshold Descriptions

6. Alerts

An alert occurs when a specific event occurs. The user is notified when the conditions for an alert are met. Alerts are generated when a scenario is run.

Alert	Description	Required
company_did_not_pay	The named company did not pay taxes by the deadline month.	Yes
company_did_pay	The named company did pay taxes by the deadline month.	No

Table #: Alert Descriptions

7. Reconciliation

Each month, the bank sends a bank statement to show the number of deposits made by the system to the Walamoo tax account. The user can reconcile that value by entering the total value deposited into the system, whereby the system will compare that value to the total taxes paid. If there is a difference, the system will indicate the difference.

8. Queries

The results of all queries should be displayed on the screen. The date, time and total records generated should be included somewhere on the screen. You can use any method for inputting the information required for the queries but note that it must be an efficient and reliable mechanism.

The user should be able to deal with two types of queries, guided and ad hoc.

8.1 Guide Queries

A guided query is one where the system asks for the required information and guides the user. The information is displayed as follows.

```
<Company Name> <Company Id>
<Street Name>, <City>, <Country>
<Total taxes paid>
```

Query	Description
List by company name.	List the records in order of company name.
List by total taxes paid.	List the records in order of taxes paid. Calculate the total per company and then generate the list.
List by country.	List the records in order of country.

Table 2: Guided Queries

8.2 Ad Hoc Queries

The ad hoc queries allow the user to choose specific query attributes. The information is displayed as follows.

```
<Company Name> <Company Id>
<Street Name>, <City>, <Country>
<Total taxes paid>
```

The **list** query creates lists of information. The format is as follows.

```
list <attribute>
```

Where <attribute> is a company, country, city or tax.

Attribute	Description
company	List the records in order of company.
country	List the records in order of country.
city	List the records in order of city.

In the case of totals, the command format is as follows.

```
total <attribute>
```

Where <attribute> is person, country, city or tax.

Attribute	Description
company	Total number of taxpayers.
country	Total number of countries.
city	Total number of cities.

tax	Total amount of taxes paid.
-----	-----------------------------

For the both commands, the user can state the number of records that should be displayed. Consider the following example.

```
list company 5
```

This query states that the results should be ordered according to the company name and only the first 5 should be displayed.